

*: Difficulty -

It is difficult to perform arithmetic operation with such a number system.

② Positional number system.

characteristics -

→ use only a few symbols called digits.

→ These symbols represent different values depending on the position they occupy in the number.

☐ The value of each digit is determined by -

→ The digit itself.

→ The position of the digit in the number.

→ The base of the number system (base = total number of digits in the number system).

*: The maximum value of a single digit is always equal to one less than the value of the base.

☐ Positional number system - classification.

✓ Binary number

✓ Decimal "

✓ Octal "

✓ Hexadecimal "

} 4 Types.

Decimal

Binary

Octal

Hexadecimal.

□ Decimal number system.

characteristics.

⇒ Has 10 symbols or digits (0, 1, 2, 3, 4, 5, 6, 7, 8, 9).

Hence, its base = 10

⇒ The maximum value of a single digit is 9 (one less than the value of the base)

⇒ We use this number system in our day to day life.

Example : $2586_{10} = (2 \times 10^3) + (5 \times 10^2) + (8 \times 10^1) + (6 \times 10^0)$
 $= 2000 + 500 + 80 + 6$
 $= 2586_{10}$

□ Binary number system.

characteristics,

⇒ Has only 2 symbols or digits (0 and 1.)

Hence, its base = 2

⇒ The maximum value of a single digit is 1

⇒ This number system is used in computers.

Binary number

" decimal

" octal

" hexadecimal

Example: 10101_2

$$\begin{aligned} &= (1 \times 2^4) + (0 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) \\ &= 16 + 0 + 4 + 0 + 1 \\ &= 21_{10} \text{ (Decimal)} \end{aligned}$$

Bit,

⇒ Bit stands for binary digit.

⇒ A bit computer terminology means either a 0 or a 1.

⇒ A binary number consisting of n bits is called an n -bit number.

Octal

Octal number system.

⇒ Has total 8 symbols or digits (0, 1, 2, 3, 4, 5, 6, 7)

⇒ Hence, its base = 8

⇒ The maximum value of a single digit is 7.

Example: 2057_8

$$\begin{aligned} &= (2 \times 8^3) + (0 \times 8^2) + (5 \times 8^1) + (7 \times 8^0) \\ &= 1024 + 0 + 40 + 7 \\ &= 1071_{10} \end{aligned}$$

Hexadecimal number system.

Has total 16 symbols or digits (0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F) Hence base = 16

A, B, C, D, E, F = 10, 11, 12, 13, 14, 15, decimal

The maximum value of a single digit is 15.

Example: $1AF_{16}$

$$= (1 \times 16^2) + (A \times 16^1) + (F \times 16^0)$$

$$= (1 \times 256 + 10 \times 16 + 15 \times 1)$$

$$= 256 + 160 + 15$$

$$= 431_{10}$$

Converting a number of Another Base to a decimal number.

Example: $4706_8 = ?_{10}$

$$4706_8 = (4 \times 8^3) + (7 \times 8^2) + (0 \times 8^1) + (6 \times 8^0)$$

$$= 4 \times 512 + 7 \times 64 + 0 + 6 \times 1$$

$$= 2048 + 448 + 6 \leftarrow \text{Sum of these products,}$$

$$= 2502_{10}$$

Decimal to Binary, Octal, Hexadecimal

* Decimal to Binary, $(49.34)_{10} = (?)_2$

$(49)_{10} = (110001)_2$ পূর্ণাংক

$\therefore (49.34)_{10} = (110001.010101)_2$

(101) পূর্ণাংক নাকি Octal অংক
তাই আমি (101)₂, (101)₈ (101)₁₆
আজ্ঞা এ ক্ষেত্রে B, D, O. অংক

4	0.34
x2	
0	.68
x2	
1	.36
x2	
0	.72
x2	
1	.44
x2	
0	.88
x2	
1	.76
	...

2	49
2	24 → 1
2	12 → 0
2	6 → 0
2	3 → 0
2	1 → 1
	0 - 1

$(0.34)_{10} = (0.010101...)_{2}$

↓ যখন 0.00 হবে তখন
আমাদের গুন এর কাজ শেষ
হবে।

* Decimal to Octal,

$(43.79)_{10} = (?)_8$ Octal convert,

পূর্ণাংক $(43)_{10}$

8	43
8	5 → 3
	0 → 5

$\therefore (43)_{10} = (53)_8$

অংশ (0.79)

0.79	
x8	
6	.42
x8	
3	.36
x8	
2	.88
x8	
7	.04
	...

$\therefore (0.79)_{10} = (0.6327...)_{8}$

$\therefore (43.79)_{10} = (53.6327...)_{8}$

* D to H $(44.69)_{10} = (?)_{16}$

16	44
16	2 → 12 (C)
	0 → 2

$\therefore (44)_{10} = (2C)_{16}$

0.69	
x16	
11	.04
x16	
0	.64
x16	
10	.24
x16	
3	.84
	...

$\therefore (0.69)_{10} = (0.0A3)_{16}$

$\therefore (44.69)_{10} = (2C.0A3)_{16}$

Q 10 to Binary $\Rightarrow$ 3 bits formula.

$$(263.715)_2 = (?)_2$$

$2^8 \rightarrow 3 \text{ bits}$

$$\frac{2}{010} \quad \frac{6}{110} \quad \frac{3}{011} \cdot \frac{7}{111} \quad \frac{1}{001} \quad \frac{5}{101}$$

$$\therefore (1011001.111001101)_2$$

	4	2	1
2	0	1	0
6	1	1	0
3	0	1	1
7	1	1	1
1	0	0	1
5	1	0	1

Q 11 to Binary $\Rightarrow$ 4 bits Formula,

$$(A29.DEF0)_{16} = (?)_2$$

$$\frac{A}{1010} \quad \frac{2}{0010} \quad \frac{9}{1001} \cdot \frac{D}{1101} \quad \frac{E}{1110} \quad \frac{F}{1111} \quad \frac{0}{0000}$$

$$= (101000101001.1101111011110000)_2$$

	8	4	2	1
A=10	1	0	1	0
B=11	0	0	1	0
C=12	1	0	0	1
D=13	1	1	0	0
E=14	1	1	1	0
F=15	1	1	1	1
0	0	0	0	0

Q Binary to Hexad, $\Rightarrow$ 4 bits Formula,

$$(101101.11010101)_2 = (?)_{16}$$

$2^{16} \rightarrow 4 \text{ bits}$

$$\begin{array}{cccc} \overleftarrow{0010} & \overrightarrow{1101} & \overrightarrow{1101} & \overrightarrow{0101} \\ 2 & 13 & 13 & 5 \end{array}$$

$$= (2D.D5)_{16}$$

Az.

	8	4	2	1
A=10	0	0	1	0
B=11	1	1	0	1
C=12	1	1	0	1
D=13	0	1	0	1
E=14				
F=15				

□ Converting a decimal number to a number of Another Base.

Example: $952_{10} = ?_8$

Soln,

$$\begin{array}{r} 8 \overline{) 952} \\ 8 \overline{) 119} \\ 8 \overline{) 14} \\ 8 \overline{) 1} \\ 0 \end{array}$$

0
7
6
1

Hence, $952_{10} = 1670_8$

$$\begin{array}{r} 2 \overline{) 10} \\ 2 \overline{) 10} \\ 0 \end{array}$$

$$\begin{array}{r} 2 \overline{) 49} (24 \\ 4 \overline{) 49} \\ 9 \\ 8 \\ 1 \end{array}$$

$$\begin{array}{r} 2 \overline{) 49} \\ 2 \overline{) 24} - 1 \\ 2 \overline{) 12} - 0 \\ 2 \overline{) 6} - 0 \\ 2 \overline{) 3} - 0 \\ 2 \overline{) 1} - 1 \end{array}$$

□ $545_6 = ?_4$

Step 1: Convert from base 6 to base 10.

$$\begin{aligned} 545_6 &= 5 \times 6^2 + 4 \times 6^1 + 5 \times 6^0 \\ &= 5 \times 36 + 4 \times 6 + 5 \times 1 \\ &= 180 + 24 + 5 \\ &= 209_{10} \end{aligned}$$

$$\begin{array}{r} 8 \overline{) 952} (119 \\ 8 \overline{) 952} \\ 119 \\ 610 \end{array}$$

Step 2: Convert 209_{10} to base 4.

$$\begin{array}{r} 4 \overline{) 209} \\ 4 \overline{) 52} \\ 4 \overline{) 13} \\ 4 \overline{) 3} \\ 0 \end{array}$$

1
0
1
3

$$4 \overline{) 209}$$

Hence, $209_{10} = 3101_4$
So, $545_6 = 209_{10} = 3101_4$ } Thus, $545_6 = 3101_4$

☐ shortcut method for converting a Binary number to its Equivalent octal number.

Example:

$$1101010_2 = ?_8$$

Step 1: Divide the binary digits into group of 3 starting from right.

$$001 \ 101 \ 010$$

Step 2: Convert each group into one octal digit

$$001_2 = 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 1$$

$$101_2 = 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 5$$

$$010_2 = 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 2$$

Hence, $1101010_2 = 152_8$

$$\begin{array}{r} 4 \ 2 \ 1 \\ 0 \ 0 \ 1 = 1 \\ 1 \ 0 \ 1 = 5 \\ 0 \ 1 \ 0 = 2 \end{array}$$

☐ shortcut method for converting an octal number to its Equivalent Binary number.

Example: $562_8 = ?_2$

Step 1: Convert each octal digit to 3 binary

$$5_8 = 101_2 \quad 6_8 = 110_2 \quad 2_8 = 010_2$$

Step 2: Combine the binary groups.

$$562_8 = \begin{array}{ccc} 101 & 110 & 010 \\ 5 & 6 & 2 \end{array}$$

Computer Arithmetic

$$\begin{array}{r}
 10000 \\
 (-) 01001 \\
 \hline
 011
 \end{array}
 \qquad
 \begin{array}{r}
 10000 \\
 (-) 01011 \\
 \hline
 00101
 \end{array}$$

Binary
mid

□ Binary Number system :

- Digits : 0 and 1.
- Bit (short for binary digit) A single binary d.
- LSB (least significant bit) The rightmost bit.
- MSB (most " bit) The Left most bit.
- Upper Byte (or nybble) The righthand byte.
- Lower Byte (or nybble) The lefthand byte.
- The term nibble used for 4 bits being a subset of byte.

□ Binary Equivalents :-

- 1 Nybble = 4 bits
- 1 Byte = 2 Nybbles = 8 bits
- 1 kilobyte = 1024 bytes.
- 1 megabyte (MB) = 1024 kilobyte = $1,048,576$ bytes
- 1 Gigabyte (GB) = 1024 (MB) = $1,073,741,824$ bytes

Important